

Администрирование сетевых подсистем

Лабораторная работа №13

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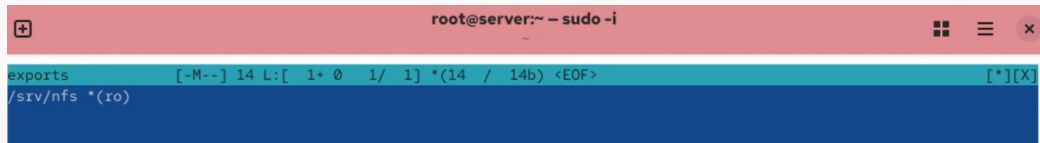
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Цели и задачи работы

Приобретение навыков настройки сервера NFS для удалённого доступа к ресурсам и организации совместного использования данных в сети.

Выполнение лабораторной работы

Настройка сервера NFSv4



A terminal window with a pink title bar. The title bar contains a window icon on the left, the text "root@server:~ - sudo -i" in the center, and window control icons (maximize, close) on the right. The terminal content shows the file "/etc/exports" being edited. The first line is "exports" followed by a status bar "[-M--] 14 L: [1+ 0 1/ 1] *(14 / 14b) <EOF> [*][X]". The second line is the export rule "/srv/nfs *(ro)".

```
root@server:~ - sudo -i
exports [ -M-- ] 14 L: [ 1+ 0 1/ 1 ] *(14 / 14b) <EOF> [*][X]
/srv/nfs *(ro)
```

Рис. 1: Редактирование файла /etc/exports

Настройка firewall и служб

```
[root@server.manovichkov.net ~]# mkdir -p /srv/nfs
[root@server.manovichkov.net ~]# mcedit /etc/exports

[root@server.manovichkov.net ~]# semanage fcontext -a -t nfs_t "/srv/nfs(/.*)?"
[root@server.manovichkov.net ~]# restorecon -vR /srv/nfs/
Relabeled /srv/nfs from unconfined_u:object_r:var_t:s0 to unconfined_u:object_r:nfs_t:s0
[root@server.manovichkov.net ~]# systemctl start nfs-server.service
[root@server.manovichkov.net ~]# systemctl enable nfs-server.service
Created symlink '/etc/systemd/system/multi-user.target.wants/nfs-server.service' → '/usr/lib/systemd/system/nfs-server.service'.
[root@server.manovichkov.net ~]# firewall-cmd --add-service=nfs
success
[root@server.manovichkov.net ~]# firewall-cmd --add-service=nfs --permanent
success
[root@server.manovichkov.net ~]# firewall-cmd --reload
success
```

Рис. 2: Настройка файервола для NFS

Проверка подключения клиента

```
Installing          : sssd-nfs-idmap-2.10.2-3.el10_0.2.x86_64      7/7
Running scriptlet: sssd-nfs-idmap-2.10.2-3.el10_0.2.x86_64      7/7

Installed:
  gssproxy-0.9.2-10.el10.x86_64      libev-4.33-14.el10.x86_64
  libnfsidmap-1:2.8.2-3.el10.x86_64  libverto-libev-0.3.2-10.el10.x86_64
  nfs-utils-1:2.8.2-3.el10.x86_64   rpcbind-1.2.7-3.el10.x86_64
  sssd-nfs-idmap-2.10.2-3.el10_0.2.x86_64

Complete!
[root@client.manovichkov.net ~]# showmount -e server.manovichkov.net
clnt_create: RPC: Unable to receive
[root@client.manovichkov.net ~]#
```

Рис. 3: Ошибка RPC при подключении

Добавление служб rpc-bind и mountd



```
root@server:~# sudo -i

hcpv6-client distcc dns dns-over-quit dns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-cl
ient etcd-server factorio finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freei
pa-trust ftp galera ganglia-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap
imaps iperf2 iperf3 ipfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana kl
ogin kpasswd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manage
r kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet kube
let-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp m
anagesieve matrix mdns memcache minecraft minidlna mndp mongodb mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mys
ql nbd nebula need-for-speed-most-wanted netbios-ns netdata-dashboard nfs nfs3 nmap-0183 nripe ntp nut opentelemetry o
penvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pncd pmpoxy pmwebapi pmwebapis pop3 pop3s postgresql
privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius r
adsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane se
tillers-history-collection sip sips slimevr slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroa
k-lansync spotify-sync squid ssdp ssh ssh-custom statrs steam-lan-transfer steam-streaming stellaris stronghold-crus
ader stun stuns submission supertuxkart svdrp svn syncthing syncthing-gui syncthing-relay synergy syscomlan syslog sy
slog-tls telnet tentacle terraria tftp tile38 tinc tor-socks transmission-client turn turns upnp-client vdsu vnc-serv
er vrrp warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-host ws-discovery-tcp
ws-discovery-udp wsdd wsdd-http wsmann xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-
java-gateway zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier

[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]# firewall-cmd --add-service=mountd --add-service=rpc-bind
success
[root@server.manovichkov.net ~]# firewall-cmd --add-service=mountd --add-service=rpc-bind --permanent
success
[root@server.manovichkov.net ~]# firewall-cmd --reload
success
[root@server.manovichkov.net ~]#
```

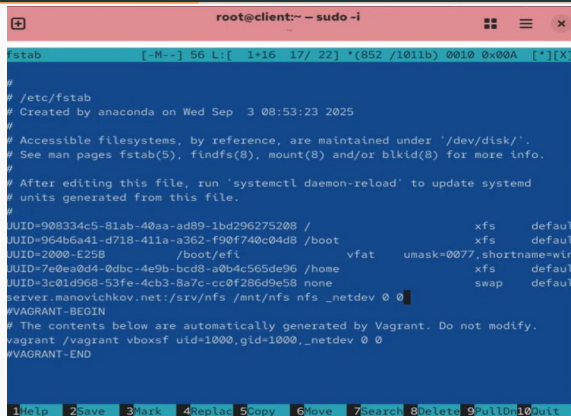
Рис. 4: Добавление служб rpc-bind и mountd


```
[root@client.manovichkov.net ~]# showmount -e server.manovichkov.net
clnt_create: RPC: Unable to receive
[root@client.manovichkov.net ~]# showmount -e server.manovichkov.net
Export list for server.manovichkov.net:
/srv/nfs *
[root@client.manovichkov.net ~]#
[root@client.manovichkov.net ~]# showmount -e server.manovichkov.net
Export list for server.manovichkov.net:
/srv/nfs *
[root@client.manovichkov.net ~]# █
```

Рис. 5: Список экспортируемых каталогов

Монтирование NFS на клиенте

Проверка монтирования



```
root@client:~ - sudo -i
fstab      [-M--] 56 L:[ 1+16 17/ 22] *(852 /1011b) 0010 0x00A [*][X]
#
# /etc/fstab
# Created by anaconda on Wed Sep  3 08:53:23 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=908334c5-81ab-40aa-ad89-1bd296275208 /                xfs     default
UUID=964b6a41-d718-411a-a362-f90f740c04d8 /boot             xfs     default
UUID=2000-E25B      /boot/efi         vfat     umask=0077,shortname=win
UUID=7e0ea0d4-0dbc-4e9b-bcd8-a0b4c565de96 /home             xfs     default
UUID=3c01d968-53fe-4cb3-8a7c-cc0f286d9e58 none              swap     default
server.manovichkov.net:/srv/nfs /mnt/nfs nfs _netdev 0 0
#VAGRANT-BEGIN
# The contents below are automatically generated by Vagrant. Do not modify.
vagrant /vagrant vboxsf uid=1000,gid=1000,_netdev 0 0
#VAGRANT-END

1Help  2Save  3Mark  4Replac 5Copy  6Move  7Search 8Delete 9PullDn10Quit
```

Рис. 6: Проверка вывода mount

```
[root@client.manovichkov.net ~]# mkdir -p /mnt/nfs
[root@client.manovichkov.net ~]# mount server.manovichkov.net:/srv/nfs /mnt/nfs
[root@client.manovichkov.net ~]# mount | grep srv
server.manovichkov.net:/srv/nfs on /mnt/nfs type nfs4 (rw,relatime,vers=4.2,rsiz
e=262144,wsiz=262144,namlen=255,hard,proto=tcp,timeo=600,retrans=2,sec=sys,clie
ntaddr=192.168.1.30,local_lock=none,addr=192.168.1.1)
[root@client.manovichkov.net ~]#
[root@client.manovichkov.net ~]# mcedit /etc/fstab

[root@client.manovichkov.net ~]#
[root@client.manovichkov.net ~]# systemctl status remote-fs.target
● remote-fs.target - Remote File Systems
   Loaded: loaded (/usr/lib/systemd/system/remote-fs.target; enabled; preset:
   Active: active since Wed 2025-10-29 07:52:27 UTC; 12min ago
   Invocation: 3c92e7bcd4ef4140918f7f99e920cc85
   Docs: man:systemd.special(7)

Oct 29 07:52:27 client.manovichkov.net systemd[1]: Reached target remote-fs.tar
```

Рис. 7: Запись в /etc/fstab

Проверка статуса remote-fs.target

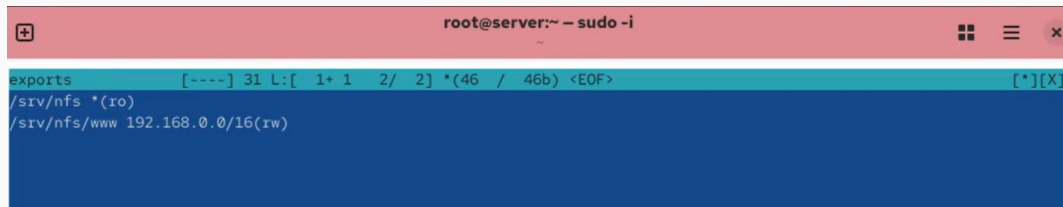
```
[root@client.manovichkov.net ~]# mkdir -p /mnt/nfs
[root@client.manovichkov.net ~]# mount server.manovichkov.net:/srv/nfs /mnt/nfs
[root@client.manovichkov.net ~]# mount | grep srv
server.manovichkov.net:/srv/nfs on /mnt/nfs type nfs4 (rw,relatime,vers=4.2,rsiz
e=262144,wsiz=262144,namlen=255,hard,proto=tcp,timeo=600,retrans=2,sec=sys,clie
ntaddr=192.168.1.30,local_lock=none,addr=192.168.1.1)
[root@client.manovichkov.net ~]#
```

Рис. 8: Проверка remote-fs.target

Подключение каталогов к дереву NFS



Создание каталога для веб-сервера

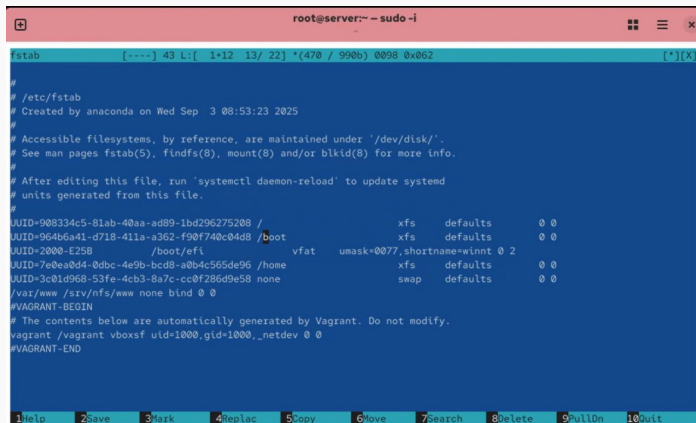


A terminal window with a pink title bar containing the text "root@server:~ - sudo -i" and window control icons. The terminal has a blue background and shows the following commands and output:

```
exports      [----] 31 L:[ 1+ 1  2/  2] *(46 / 46b) <EOF> [*][X]  
/srv/nfs *(ro)  
/srv/nfs/www 192.168.0.0/16(rw)
```

Рис. 9: Создание /srv/nfs/www

Монтирование каталога веб-сервера



```
root@server:~ -- sudo -i
fstab [-----] 43 L: [ 1*12 13/ 22] *(470 / 990b) 0098 0x062 [*][X]

#
# /etc/fstab
# Created by anaconda on Wed Sep  3 08:53:23 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=908334c5-81ab-40aa-ad89-1bd296275208 / xfs defaults 0 0
UUID=964b6a41-d718-411a-a362-f90f740c04d8 /boot xfs defaults 0 0
UUID=2000-E258 /boot/efi vfat umask=0077,shortname=winnt 0 2
UUID=7e0ea0d4-0dbc-4e9b-bcd8-a0b4c565de96 /home xfs defaults 0 0
UUID=3c01d968-53fe-4cb3-8a7c-cc0f286d9e58 none swap defaults 0 0
/var/www /srv/nfs/www none bind 0 0
#VAGRANT-BEGIN
# The contents below are automatically generated by Vagrant. Do not modify.
vagrant /vagrant vboxsf uid=1000,gid=1000,_netdev 0 0
#VAGRANT-END

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рис. 10: Монтирование /var/www в /srv/nfs/www


```
[root@server.manovichkov.net ~]# mkdir -p /srv/nfs/www
[root@server.manovichkov.net ~]# mount -o bind /var/www /srv/nfs/www
[root@server.manovichkov.net ~]# ls /srv/nfs
www
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]# mcedit /etc/exports

[root@server.manovichkov.net ~]# exportfs -r
```

Рис. 11: Редактирование /etc/exports

```
[root@client.manovichkov.net ~]# ls /mnt/nfs/  
www  
[root@client.manovichkov.net ~]# ls /mnt/nfs/  
www  
[root@client.manovichkov.net ~]# ls /mnt/nfs/  
www  
[root@client.manovichkov.net ~]# █
```

Рис. 12: Проверка каталога /mnt/nfs

Подключение каталогов для пользователей



Создание пользовательского каталога

```
[manovichkov@server.manovichkov.net ~]$  
[manovichkov@server.manovichkov.net ~]$ mkdir -p -m 700 ~/common  
[manovichkov@server.manovichkov.net ~]$ cd ~/common/  
[manovichkov@server.manovichkov.net common]$ touch manovichkov@server.txt  
[manovichkov@server.manovichkov.net common]$ ls  
manovichkov@server.txt  
[manovichkov@server.manovichkov.net common]$  
[manovichkov@server.manovichkov.net common]$ ls  
manovichkov@client.txt  manovichkov@server.txt
```

Рис. 13: Создание каталога и файла пользователя

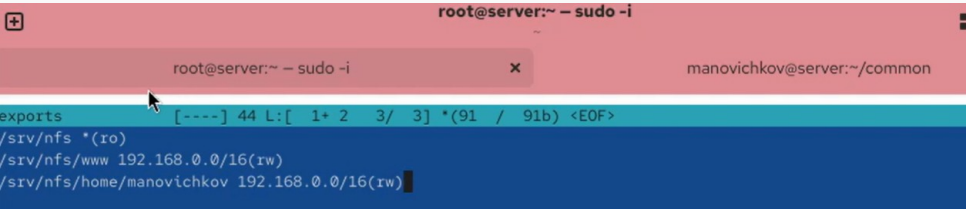
Создание каталога для сетевого доступа

```
[root@server.manovichkov.net ~]# mkdir -p /srv/nfs/home/manovichkov
[root@server.manovichkov.net ~]# mount -o bind /home/manovichkov/common/ /srv/nfs/home/manovichkov/
mount: (hint) your fstab has been modified, but systemd still uses
        the old version; use 'systemctl daemon-reload' to reload.
[root@server.manovichkov.net ~]#
[root@server.manovichkov.net ~]# mcedit /etc/exports

[root@server.manovichkov.net ~]# exportfs -r
[root@server.manovichkov.net ~]# mcedit /etc/fstab

[root@server.manovichkov.net ~]# exportfs -r
```

Рис. 14: Создание /srv/nfs/home/user

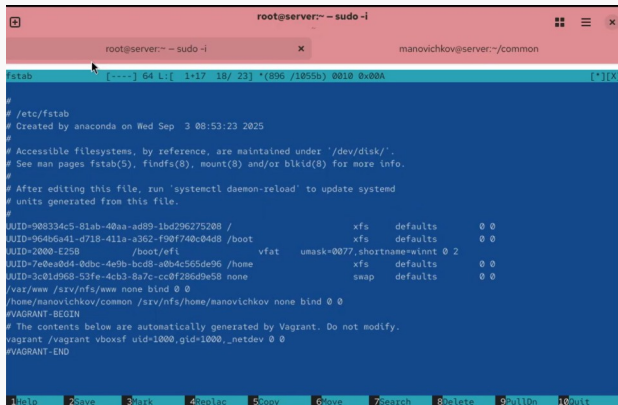


The screenshot shows a terminal window with a pink title bar. The title bar contains a window control icon on the left, the text "root@server:~ - sudo -i" in the center, and a close button on the right. Below the title bar, there are two tabs: "root@server:~ - sudo -i" (active) and "manovichkov@server:~/common". The terminal content shows the file "/etc/exports" with the following lines: "[----] 44 L:[1+ 2 3/ 3] *(91 / 91b) <EOF>", "/srv/nfs *(ro)", "/srv/nfs/www 192.168.0.0/16(rw)", and "/srv/nfs/home/manovichkov 192.168.0.0/16(rw)". A mouse cursor is pointing at the first line of the file content.

```
root@server:~ - sudo -i
root@server:~ - sudo -i x manovichkov@server:~/common
exports [----] 44 L:[ 1+ 2 3/ 3] *(91 / 91b) <EOF>
/srv/nfs *(ro)
/srv/nfs/www 192.168.0.0/16(rw)
/srv/nfs/home/manovichkov 192.168.0.0/16(rw)
```

Рис. 15: Добавление каталога в /etc/exports

Автоматическое монтирование пользователя



```
root@server:~ - sudo -i
root@server:~ - sudo -i
manovichkov@server:~/common

fstab [----] 64 L: [ 1+17 18/ 23] *(896 /1055b) 0010 0x00A [*][X]

#
# /etc/fstab
# Created by anaconda on Wed Sep  3 08:53:23 2025
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=908334c5-81ab-40aa-ad89-1bd296275208 / xfs defaults 0 0
UUID=964b6a41-d718-411a-a362-f90f740c04d8 /boot xfs defaults 0 0
UUID=2000-E258 /boot/efi vfat umask=0077,shortname=winnt 0 2
UUID=7e0ea0d4-0dbc-4e9b-bcd8-a0b4c565de96 /home xfs defaults 0 0
UUID=3c01d968-53fe-4cb3-8a7c-cc0f286d9e58 none swap defaults 0 0
/var/www /srv/nfs/www none bind 0 0
/home/manovichkov/common /srv/nfs/home/manovichkov none bind 0 0
#VAGRANT-BEGIN
# The contents below are automatically generated by Vagrant. Do not modify.
vagrant /vagrant vboxsf uid=1000,gid=1000,_netdev 0 0
#VAGRANT-END

1Help 2Save 3Mark 4Replac 5Copy 6Move 7Search 8Delete 9PullDn 10Quit
```

Рис. 16: Редактирование /etc/fstab

```
[manovichkov@client.manovichkov.net ~]$ cd /mnt/nfs/home/manovichkov/  
[manovichkov@client.manovichkov.net manovichkov]$ ls  
manovichkov@server.txt  
[manovichkov@client.manovichkov.net manovichkov]$ touch manovichkov@client.txt  
[manovichkov@client.manovichkov.net manovichkov]$ ls  
manovichkov@client.txt  manovichkov@server.txt
```

Рис. 17: Создание файла на клиенте

Проверка синхронизации на сервере

```
[manovichkov@server.manovichkov.net ~]$ mkdir -p -m 700 ~/common
[manovichkov@server.manovichkov.net ~]$ cd ~/common/
[manovichkov@server.manovichkov.net common]$ touch manovichkov@server.txt
[manovichkov@server.manovichkov.net common]$ ls
manovichkov@server.txt
[manovichkov@server.manovichkov.net common]$
[manovichkov@server.manovichkov.net common]$ ls
manovichkov@client.txt  manovichkov@server.txt
```

Рис. 18: Появление изменений на сервере

Выводы по проделанной работе

В ходе лабораторной работы был установлен и настроен сервер NFSv4. Реализовано предоставление общих каталогов, выполнено их монтирование на клиенте, а также организована работа пользовательских каталогов через NFS. Настроено автоматическое подключение при загрузке системы и создан скрипт автоматизации конфигурации в виртуальной среде.